# EBNUL HASAN MAHI

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## PROFESSIONAL SUMMARY

Software Engineering student (CGPA: 3.67) with practical experience building responsive full-stack web applications, secure authentication workflows, and relational/document database schemas. Proficient in React, Next.js, Node.js, Express, MySQL, and WebSockets. Committed to clean architecture, reliable engineering fundamentals, and practical software design.

## TECHNICAL SKILLS

**Languages:** JavaScript (ES6+), TypeScript, Java, Python, C/C++, SQL, HTML5, CSS3
**Frontend:** React.js, Next.js, Tailwind CSS, Responsive Design, Web Accessibility
**Backend & Auth:** Node.js, Express.js, Spring Boot, FastAPI, JWT, REST APIs, WebSockets
**Databases:** MySQL, PostgreSQL, MongoDB
**Tools & Concepts:** Git, GitHub, Postman, Vite, Vercel, Docker, OOP, Data Structures & Algorithms

## FEATURED PROJECTS

**Dr. Md. Momenul Islam Health Assistant | *React, TypeScript, FastAPI, Python, RAG* — Prototype | GitHub**
*Evidence-grounded health information assistant prototype — Built in memory of Dr. Md. Momenul Islam*

- Developed a multilingual health assistant interface addressing queries across English, Bangla, and Banglish.
- Grounded clinical guidance using curated NHS sources with transparent source provenance tracking and safety disclaimers.
- Implemented deterministic out-of-scope refusal protocols and emergency routing to avoid ungrounded generative advice.
- Built modern web interface; retrieval pipeline runs in local high-memory environment with full triage walkthrough.

**NoboGhat | *Next.js, React, Java, Spring Boot, MySQL, JWT* — Live Demo | GitHub**
*Decoupled inland waterway logistics & cargo booking platform*

- Developed a responsive, accessible frontend using Next.js and Tailwind CSS with server-side rendering.
- Engineered decoupled backend services with Java, Spring Boot, and MySQL for boat discovery and capacity-aware booking.
- Implemented live cargo capacity tracking and role-based access control for farmers, traders, and boat operators.

**HeliosTrack | *React, Node.js, Express, MySQL, WebSockets, Chart.js* — Live Demo | GitHub**
*Real-time distributed solar array monitoring dashboard with sub-second telemetry*

- Designed and built an enterprise telemetry dashboard featuring sub-second live telemetry streaming via WebSockets.
- Integrated interactive Chart.js grid and virtual battery controller to visualize distributed solar array performance.
- Connected Node.js API with Aiven Cloud MySQL database for reliable persistence and fast querying of inverter metrics.

## EDUCATION & TRAINING

**Bachelor of Science in Software Engineering — Daffodil International University (DIU)**
*Current CGPA: 3.67 | Expected Graduation: 2028*

**Full-Stack Web Development Training — Programming Hero (Batch 14, In Progress)**
*Intensive practical training covering modern JavaScript, React, Next.js, Node.js, Express, MySQL, and Web Architecture.*